

1. Eccentricity vector meaning.

$$\underline{e} = \frac{\underline{v} \times \underline{h}}{\mu} - \frac{\underline{r}}{r}$$

$$e = |\underline{e}|$$

이성궤를 벗어나 양행은 근궤 양행, 크기는 이심률.

why dimensionless?

$$\underline{h} = \underline{r} \times \underline{v}$$

$$r = [\text{km}]$$

$$v = [\text{km/s}]$$

$$h = [\text{km}^2/\text{s}]$$

$$\underline{v} \times \underline{h} = [\text{km}^3/\text{s}^2]$$

$$\mu = [\text{km}^3/\text{s}^2]$$

$$\frac{\underline{v} \times \underline{h}}{\mu} = \frac{\text{km}^3/\text{s}^2}{\text{km}^3/\text{s}^2} \longrightarrow \text{Non-D}$$

2. Energy to semi-major axis

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r}$$

$$\epsilon = -\frac{\mu}{2a}$$

$$a = -\frac{\mu}{2\epsilon}$$

if $\epsilon < 0$, $a > 0 \longrightarrow$ Bound elliptical orbit
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